

'Micropipeline.'

This protocol seems very efficient in both time and energy because of its very little switching activity. But components sensitive on transitions are more complex than elements, which just react to signal levels. In fact, this protocol is very good where high speed is preferred over energy or space efficiency.

Four-phase bundled-data protocol. The four-phase bundled-data protocol, also known as four-phase single-rail protocol, is very similar to the two-phase bundled-data protocol. So only the differences between the two are pointed out here.

This protocol has more switching activity than the two-phase protocol, which may lead to slower and more energy consuming circuits, but implementations sensitive to transitions are often more complex than those sensitive to levels.

The Muller-C element

The Muller-C element is fundamental for asynchronous pipelines. It is an essential part of the Muller pipeline. Its function may be described as the following 'while' loop:

```
while true do
  if x == y then
    z = x
  else
    z = z
  end if
done
```

In other words, the Muller-C element changes its output value only if both inputs have the same value. Otherwise, it retains its value. Using this element together with the handshaking methods explained earlier leads to a very fundamental pipeline technique (Muller pipeline) that solves the problem of propagating only the valid data.

Properties of asynchronous design

Asynchronous processors have many advantages over their synchronous counterparts because the central clock

is abandoned.

Self-timed. Communication between several parts of the processor can be self-timed. This means that the correct logic operation is independent of propagation delays within specific operating constraints. This is achieved by using handshaking methods such as those described earlier. Because of this, the receiver knows whether valid data is given at its input ports and the sender knows whether the receiver has captured the data. If there are delays in signal propagation, the circuit state is still valid.

Energy efficiency. In synchronous processors, the clock itself consumes a significant amount of energy (dynamic power usage), which can be up to 40 per cent of the overall power consumption. If an asynchronous processor is idle, it wastes static energy (leakage energy), which is very low. But this advantage is getting smaller with shrinking structure sizes, which causes the static power consumption to increase. Energy efficiency of asynchronous processors is also based on the fact that operational speed is always as fast as required until the maximum is reached.

Maximum speed operation. Since asynchronous processors do not have a central clock, there is no need to adjust operational speed of each part. Instead, each part can operate at its currently operation-dependent maximum speed without affecting the other components' speed. Imagine an arithmetic logical unit (ALU) that can execute an add operation in half the time of a multiplication. In synchronous processors, the clock cycle has to be at least as long as the slowest operation (critical path) even if this operation is rather rare. In asynchronous processors, the system is slowed down only if this rare operation occurs—else, it operates faster.

Utilisation of processors is very low for some applications and energy is wasted without any return. Modern synchronous CPUs like Transmeta's LongRun try to accommodate this by

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